

Optimal Tone Melodies in Shanghainese: a unified OT account of word-level tone sandhi

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Abstract

This paper presents a unified Optimality Theoretic analysis of left-dominant word-level tone sandhi in Shanghainese. The proposal consists of a single constraint hierarchy capable of accounting for the disparate sandhi behavior initiated by the language's five lexical tones. By appealing to one basic faithfulness condition and one correspondence constraint that formalizes a familiar rightward shifting pattern, the analysis can account for the major tone melodies induced by each of the five tones—from disyllables to tetrasyllabic expressions—with five additional markedness conditions that regulate only tone melodies in general, including an avoidance of adjacent High tones and an avoidance of melody reversal. One additional contour-faithfulness constraint extends the grammar's coverage to accurately predict the monosyllabic citation forms for each tone from its underlying representation.

Keywords: Shanghainese, tone sandhi, tone melodies, left-dominant tone sandhi, left-prominent tone sandhi, word-level tone sandhi, tone shifting, tone spreading, Wu languages, Sinitic tone systems, Optimality Theory

1 Introduction

This paper proposes an Optimality Theory (OT) analysis of Shanghainese word-level tone sandhi. Shanghainese boasts an intricate system of word-level tone melodies that are 'left-dominant' in that the tone pattern over a polysyllabic expression is determined by the tone of the first syllable. We begin with an overview of the five tones of Shanghainese and the specific tone melodies induced by each of them across two-, three-, and four-syllable expressions in which they appear initially.

Section 3 is a closer look at the sandhi behavior of each tone class and a review of other tone-based phonological accounts of the melodies. Based on their sandhi behavior, the five tones sort into three groups: two 'white sheep,' T2 and T3, whose melodies follow a basic mapping pattern; two 'gray sheep,' T1 and T4, which partially depart from that pattern in different ways; and a lone 'black sheep,' T5, whose behavior has long been regarded as disconnected from the rest and does not fit the pattern at all. One of the conclusions of this paper is that the black sheep is not what it appears. T5, routinely singled out as the most enigmatic of the five classes, turns out on our account to be the most straightforward of them all. The reason, we argue, is that the sandhi system turns on an answer to a single question: *where, in a tone's underlying representation, is the High element?* A melody initiated by a tone with no High element (T5) stays flat until a final contour is forced upon it. A melody initiated by a tone with a High element on its right edge

(T2, T3, T4) shifts the High element rightward onto the second syllable; then, having peaked, must descend and terminate with the signature final contour. (T4 appears unusual since it has only one element, but the initial syllables of its melodies are filled with a default Mid tone.) A tone with a High tone on its left edge (T1) descends immediately, without any shifting. Thus, what look like four disparate sandhi patterns are just four answers to one question.

In section 4, we give an OT-based analysis that predicts the attested tone melodies. The account is a unified analysis of the major sandhi patterns for all five tone classes. It involves one faithfulness constraint regulating input-output correspondence, one relative markedness hierarchy of the three basic tone elements (High > Low > Mid), a correspondence constraint to account for the existence of rightward tone shifting, and four additional markedness constraints that moderate the shape and features of tone melodies in general. Section 5 shows how the grammar can predict the monosyllabic citation forms themselves with the help of one additional faithfulness constraint. The penultimate section elaborates on the analysis of underlying forms and offers an explanation for a restricted alternative T5 tetrasyllabic pattern, which we analyze as a case of diachronic analogical development.

2 Background: Shanghainese Tones and Melodies

2.1 Shanghainese tones

Early Middle Chinese (circa 600 to 1000 CE) had four tones. In Late Middle Chinese (circa 1000 to 1300 CE), each tone split into two distinct registers depending on the nature of the onset, resulting in eight total tone classes. Modern Sinitic languages preserve reflexes of these historical classes to varying degrees, with some categories merging and others undergoing further splits in different languages (Norman, 1988; Shen, 2020). Shanghainese still had eight distinct tones in the mid-nineteenth century (Edkins, 1853), but these have since merged to leave only five, often referred to as T1–T5. The lexical distribution of the five classes is tightly conditioned by onset type and syllable structure. Only two of the tones, T1 and T2, can actually distinguish words since they can both occur in open or nasal-closed syllables with a voiceless onset (cf. t_0^{T1} ‘knife’, t_0^{T2} ‘island’). The other three can occur only in syllabic structures that are unavailable to the other tones. There is only one possible obstruent coda in Shanghainese, namely /ʔ/, and tones that can appear with such a coda are commonly referred to as ‘checked’ tones. The distribution of the tones in terms of the syllabic environments in which they occur, as well as their high versus low and rising versus falling status is summarized in Table 1.

Syllable structure		/(C)V(N)/		/ (C)Vʔ/
Register	Onset	Falling	Rising	Checked
High	[–voice]	T1	T2	T4
Low	[+voice]	–	T3	T5

Table 1: Five tones of Shanghainese sorted by register, pitch contour, and syllable structure (adapted from Ye and Boersma 2024).

Traditional Chinese dialectological accounts (e.g., JSS 1960; Xu and Tang, 1988) usually describe tones using Chao’s (1930) five-point scale, with 1 representing the lowest pitch and 5 being the highest. Generative analyses (e.g., Zee and Maddieson, 1979, henceforth Z&M) typically assume a reduced tonal inventory from which finer distinctions can be derived and

use the simpler H, L, M (High, Mid, and Low) notation. Table 2 compares the surface forms assumed for the present analysis with descriptive and acoustic characterizations reported in the literature.

	T1	T2	T3	T4	T5
Present analysis	HM	MH	LM	H	LM
Ling (2025) (F_0)	53	34	13	55	23
Zee (1988) (F_0)	HL (51)	MM [†] (34)	LM [†] (14)	M [†] (44)	LM (13)
Z&M (1979) (F_0)	HL (51)	MM [†] (34)	LM [†] (14)	H (55)	LM [†] (14)
Xu & Tang (1988)	53	34	13	55	12
Norman (1988)	42	35	24	55	23
Walton (1983)	53	34	14	5	2
JSS 1960 (1960)	53	34	13 / 14	5	2

Table 2: Selected descriptions of the Shanghai tone inventory. Sources noted with (F_0) report characterizations based on acoustic measurements.

2.2 Left-dominant sandhi and tone melodies in Shanghainese

Tone sandhi is a process whereby tones undergo substantial alternation due to the influence of neighboring sounds or words in connected speech. Rose (1990, p. 1, citing Chao 1967; Wang 1967; Ballard 1980) describes Wu languages as having “the most complex tone sandhi of the various Chinese languages, and possibly of any of the world’s tone languages.” Northern Wu languages such as Shanghainese exhibit two principal types of tone sandhi: word-level tone sandhi and phrasal tone sandhi. Word-level tone sandhi in Shanghainese is left-dominant: the lexical tone of the initial syllable determines the tonal melody realized across the sandhi domain, which may consist of a word or a short, tightly connected expression such as a compound. Consequently, the underlying tones of non-initial syllables are largely or entirely irrelevant to the surface tonal pattern of the domain.¹

Table 3 presents the tone melodies associated with each initial tone class in disyllabic through tetrasyllabic domains, along with a reminder about the traditional phonetic descriptions of the citation tone. We include as well our own starting assumptions about the underlying representations of T1–T5, which we discuss in detail later in § 6.

Because the melodies of these polysyllables are solely determined by the tone class of their initial syllable, we will use terms like ‘T1 polysyllables’ or ‘the T3 disyllable’ as shorthand throughout this paper; the reader should understand that we just mean a melody that begins with the particular syllable mentioned (rather than an actual string of consecutive syllables all of that type).

¹Sandhi-domain boundaries need not coincide exactly with lexical-word boundaries. E.g., the four-syllable ǝ-da-li-ia ‘Australia’ can surface with two disyllabic domains or a single domain (Chen, 2000, 308–311); see also Duanmu (1993, 1995). The present analysis will make no attempt to predict where domain boundaries occur and is exclusively focused on the internal characteristics of the domains themselves.

Tone	X&T	UR*	2 σ	3 σ	4 σ
T1	53	/HL/	<i>H-ML</i>	<i>H-M-ML</i>	<i>H-M-M-ML</i>
T2	34	/MH/	<i>M-H</i>	<i>M-H-ML</i>	<i>M-H-M-ML</i>
T3	13	/LH/	<i>L-H</i>	<i>L-H-ML</i>	<i>L-H-M-ML</i>
T4	55	/H/	<i>M-H</i>	<i>M-H-ML</i>	<i>M-H-M-ML</i>
T5	12	/LM/	<i>L-LM</i>	<i>L-L-LM</i>	$\left\{ \begin{array}{l} L-L-L-LM \\ L-H-M-ML \end{array} \right.$

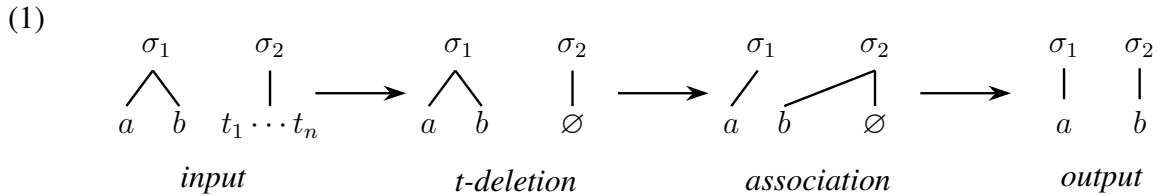
Table 3: Left-dominant sandhi tone melodies for T1–T5 in disyllabic through tetrasyllabic domains (adapted from Xu and Tang 1988, p. 24). *URs are our own working assumptions.

3 Previous Analyses

3.1 Introduction: disyllabic tone mapping and tone melodies

Shanghainese tone sandhi has received substantial attention in the generative linguistics literature (see references herein). Because left-dominant sandhi in Shanghainese involves a domain-initial element that shares its tonal characteristics with the rest of the domain (whose tones have either disappeared or been made available for overwriting), the first principal question in analyzing the phenomenon concerns how the tonal material associated with the initial syllable is distributed across subsequent syllables. The relation has most commonly been referred to as ‘spreading’ in the literature on Shanghainese (e.g., Ballard, 1980; Zhu, 1995), but whereas spreading usually implies a one-to-many mapping (Goldsmith, 1976; Williams, 1976; Pulleyblank, 1983), the most manifest pattern in Shanghainese is more of a one-to-one mapping whereby individual tonal targets associated with the initial syllable are realized successively on the first two syllables of the sandhi domain.

Zee and Maddieson (1979) (Z&M) proposed the first explicit autosegmental account that Duanmu (1999, p. 2) later called the “traditional analysis” adopted by him and many others (Yip, 1980; Lu, 1987; Selkirk and Shen, 1990; Toda, 1990; Duanmu, 1995; Zhu, 1995). Their derivation first deletes the tones of non-initial syllables and reassociates the surviving tonal material of the initial syllable with the (now) toneless syllables in the domain in order to fulfill a well-formedness condition that disallows syllables without tones.



Traditional autosegmental analysis of rightward tone re-association in a disyllabic domain (adapted from Zee and Maddieson 1979, p. 121–123).

Below we elaborate on how Shanghainese’s T2 and T3 exemplify this pattern, with their respective disyllables being easily derived from their commonly presumed underlying forms. Analysis of T4 faces a complication; its associated polysyllables (which are identical to those of T2) necessitate either a separate analysis altogether or an explanation of how its underlying form differs so pointedly from its citation form, which is a short high tone with no obvious contour

and no obvious resemblance to T2. T1 deviates in a different way from the paradigm, since its disyllable seems to follow the general rule only three-fourths of the way. T5 is the real albatross, and is generally considered an exception to the typical rightward association pattern that follows its own rule(s) for its own reason(s).

Since most analyses view the initial-syllable melody as comprising at most two distinct tonal targets, the mapping relation illustrated in (1) directly determines only the tones of the first two syllables of a domain. As such, the second principal question at hand is how the melody is extended when the sandhi domain contains more than two syllables. In this respect, T1–T4 are uniform in their behavior; when their domains are three or more syllables, their melodies all continue and terminate in the same way. Again, however, T5 exhibits a completely different pattern.

3.2 The simple cases: T2 and T3

Disyllables beginning with T2 and T3 are the straightforward cases: the two tonal targets associated with the initial syllable are realized successively across the first two syllables of the sandhi domain. Table 4 shows representative T2- and T3-initial melodies.²

Phrase	Underlying form	Surface melody	Gloss
进口	tɕin ^{MH} kʰɿ ^{MH}	<i>M–H</i>	‘import’
写下来	ɕia ^{MH} fio ^{LH} le ^{LH}	<i>M–H–ML</i>	‘write down’
半夜三更	pø ^{MH} fia ^{LH} se ^{HL} kã ^{HL}	<i>M–H–M–ML</i>	‘in the middle of the night’
牛奶	nʲiɿ ^{LH} na ^{LH}	<i>L–H</i>	‘milk’
黄霉天	fuũ ^{LH} me ^{LH} tʰi ^{HL}	<i>L–H–ML</i>	‘plum-rain season’
搞七念三	gɔ ^{LH} tɕʰiɿ ^H nʲie ^{LH} se ^{HL}	<i>L–H–M–ML</i>	‘mess around’

Table 4: Examples of T2- and T3-initial left-dominant tone sandhi.

The first two syllables of each melody follow the schema in (1); a citation contour *ab* corresponds to a pair of syllables with *a* on the first syllable and *b* on the second. (The *M–ML* sequence that shows up consistently when the sandhi domain is larger than a disyllable is something we will return to shortly.) The remaining tone classes pose challenges to the basic generalization.

3.3 The T4 problem: shifting, not spreading

T4 presents the most immediate challenge. Its surface melodies, shown in Table 5, are identical to those of T2.

These are not surface forms we would expect under the straightforward mapping schema in (1) if we assume, following Z&M, that T4 is underlyingly /H/. Under that assumption, because H is both a right edge and a left edge, both the σ_1 and σ_2 should bear the associated H-tone. That would indeed be a textbook case of tone spreading. Z&M in fact make exactly that conclusion (*Ibid.* p. 122): they claim that T4-initial disyllables do not contain a Mid tone and that both syllables bear only a High tone. However, that prediction is not supported by data cited

²The particular examples in Tables 4–7 are taken from Gu et al. (2004).

Phrase	Underlying form	Surface melody	Gloss
发展	faɿ ^H tsø ^{MH}	<i>M-H</i>	‘development’
吃生活	tɕ ^h ɿɿ ^H sã ^{HL} ɦuəɿ ^{LM}	<i>M-H-ML</i>	‘get beaten’
一塌糊涂	ɿɿ ^H t ^h aɿ ^H ɦu ^{LH} du ^{LH}	<i>M-H-M-ML</i>	‘a complete mess’

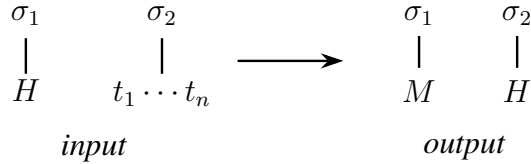
Table 5: Examples of T4-initial left-dominant tone sandhi.

by any other source that we are aware of. For detailed acoustic comparison of disyllabic tone combinations that supports the consensus that the T4-disyllable is *M-H*, see Zhang and Meng (2016, p. 177).

As noted in Table 2 in the previous section, some traditional grammars (e.g., Xu and Tang 1988) do give phonetic descriptions of T4 as having two tonal elements, but those elements are always the flat ‘44’ or ‘55’, or the falling ‘54’ contour, none of which would help to explain why T4-initial melodies begin with a Mid tone and rise to High on the second syllable, since that surface form and the others in Table 5 could not have arisen through the same simple two-target correspondence that generated the same melodies for T2.

A related observation is made by Zhu (1998), whose phonetic examination of T4+X leads him to conclude that its F_0 contour “cannot at all be considered F_0 spreading” of citation T4. He instead analyzes the pattern as ‘contour-element spreading,’ phonetically manifested as a rightward contour shift. The details of Zhu’s analysis are beyond the present scope, but the resulting disyllable is 33+44, equivalent to *M-H* in our notation.

(2)



Apparent H-tone shifting in T4-initial disyllables.

Another approach to the puzzle is that of Selkirk and Shen (1990), who follow the suggestion of Jin (1986) (citing arguments of Yip (1980) and Wright (1983, p. 161)) that “although [T4] is phonetically higher than [T2] in isolation, they show the same sandhi, and should thus have the same representation” (Jin, 1986, p. 27). If one is not satisfied with this reasoning, the pattern in (2) still leaves open the questions of why T4 does not surface as *H-H* and how the mysterious M tone shows up on the σ_1 of all T4 polysyllables, creating a pattern that is exactly the same as the one induced by an initial T2. If one does accept the proposal that the underlying form of T4 is /MH/, then there is still the problem of explaining why its citation form has no M-tone.

3.4 A caveat: questions about larger domains

So far, we have postponed discussion of the second principal question noted above. A simple two-target correspondence does nothing to explain why the additional syllables are realized as medial *M* or why the melodies systematically terminate as *ML*.

According to Zhang (2007, 262), in a T2-initial tetrasyllable like pø^M ɦia^H se^M kã^L (‘in the middle of the night’), “a boundary L is realized on the last syllable, and a Mid tone obtains on the third syllable as the result of interpolation between the second-syllable H and the right-boundary L.” This has similarities to and differences with Z&M’s (1979, 120) ‘L-Insertion’

rule that applies before spreading, which was meant to address precisely the question under consideration. In Z&M’s original proposal, L is inserted onto σ_3 after all tones except those of the initial, dominant syllable are deleted. The Low tone can then itself spread rightward in order to satisfy a separate well-formedness condition that bans toneless syllables. (See derivation in Figure 1.) Zee (1988, p. 341) subsequently refers to the Low tone as the ‘default tone’ in Shanghainese.

input	半 pø /MH/	夜 fia /LH/	三 se /HL/	更 kã /HL/
tone deletion	<i>MH</i>	∅	∅	∅
L-insertion	<i>MH</i>	∅	<i>L</i>	∅
rightward spreading	<i>M</i>	<i>H</i>	<i>L</i>	<i>L</i>

Figure 1: Autosegmental derivation of 半夜三更 (‘in the middle of the night’), adapted from Zee and Maddieson (1979, 125). The prediction of successive Low tones is incorrect, according to most sources.

The analysis of Z&M is elegant in that the same spreading step handles two portions of the melody at once. However, it is not consistent with most versions of the data nor with Zhang’s (2007) observation that the Low tone in fact appears only at the end of the melody rather than right after the second syllable, and that the tonal targets leading up to the final one are Mid tones, not Low tones. Z&M claim that “because of co-articulation effects between H on the second syllable to the following L, the pitch of the third syllable is closer to a phonetic Mid level ... this kind of coarticulation between adjacent tones is a straightforward phonetic effect” and thus requires no phonological explanation (Z&M, p. 125). However, a recent detailed examination by Takahashi (2019a) of these pitch targets supports the conclusion that the final L in these examples is indeed a boundary tone and that the tones leading up to it are M-tones both phonetically and phonologically.

Thus, contrary to Zee’s claim, if anything behaves like a default tone in Shanghainese tone melodies, it appears to be the Mid tone rather than the Low tone.³ Zhang’s remark that the intervening Mid tones appear to behave as some sort of “interpolation” between the final Low tone and the tonal targets on the second syllable is a critical empirical detail that will be relevant to what we have to say in the next section.

3.5 The T1 problem: the mystery of the premature boundary tone

Trisyllables and tetrasyllables initiated by T1 end the same way as the ones that begin with T2, T3, and T4. Oddly though, T1-initial disyllables end that way too. The ‘boundary L-tone’ just discussed seems to show up early. Representative forms are shown in Table 6.

We have seen that the ‘normal’ recipe for making Shanghainese polysyllables from dominant-tone ingredients is (i) put the left-edge tone on the first syllable, (ii) put the right-edge tone on the second syllable, and (iii) make everything else an M-tone until you reach the ‘boundary-L.’ We noted that T4 seems to skip step (i), but for a tone whose left and right edges are the same thing, this might not come as a complete shock.

But T1 skips the second step. To quote Zhang (2007, 262) again, “[t]he behavior of the [T1] tone may require extra machinery to account for, as the L from the initial syllable and the

³See Lu (1987, p. 109) for an early precursor to the idea that M is the unmarked tonal specification.

Phrase	Underlying form	Surface melody	Gloss
空调	k ^h oŋ ^{HL} diɔ ^{LH}	<i>H–ML</i>	‘air conditioning’
边浪向	pi ^{HL} lã ^{LH} ɕiã ^{MH}	<i>H–M–ML</i>	‘at the side’
三日两头	se ^{HL} n̄jiɿ ^{LM} liã ^{LH} dʑ ^{LH}	<i>H–M–M–ML</i>	‘every few days’

Table 6: Examples of T1-initial left-dominant tone sandhi.

boundary L seem to have merged into one and docked onto the right edge.” In other words, rather than forming a disyllable with a simple sequence of two level syllables corresponding transparently to the two input targets like T2–T4 do, T1 seems more eager to terminate in a way that mimics the shape of the final syllable of the longer melodies we just discussed. The question is why. Zhang hints that the answer is related to the fact that the right edge of T1’s underlying form is identical to the boundary L in longer melodies. In our analysis in the next section, we will argue that Zhang’s remark is, in a broad sense, half right and half wrong, and that the correct part of his intuition can be captured quite nicely without any complicated assumptions.

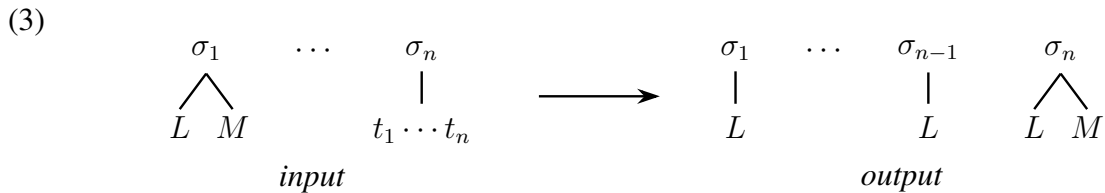
3.6 The T5 problem: spreading? displacement?

Finally, T5 presents the most striking departure from the patterns exhibited by T1–T4. Representative examples are given in Table 7.

Phrase	Underlying form	Surface melody	Gloss
蹩脚	biɿ ^{LM} tɕiaɿ ^H	<i>L–LM</i>	‘shoddy; poor’
录音机	loɿ ^{LM} in ^{HL} tɕi ^{HL}	<i>L–L–LM</i>	‘tape recorder’
复旦大学	voɿ ^{LM} te ^{MH} da ^{LH} fioɿ ^{LM}	<i>L–L–L–LM</i>	‘Fudan University’

Table 7: Examples of T5-initial left-dominant tone sandhi.

At first glance, these forms suggest an operation quite different from the one illustrated in (1). In each word, the rising contour associated with the T5 citation syllable shows up at the opposite edge of each word, while all the syllables preceding the final one carry an L-tone. The disyllabic, trisyllabic, and tetrasyllabic forms are therefore compatible with at least two representational analyses: (a) the M shifts to the right edge and the initial L spreads rightward to everything before that, or (b) the entire initial LM contour is shifted to the right edge, while all preceding syllables surface with L.



The unusual pattern of T5. The initial *LM* contour is realized at the right edge of the domain, while *L* appears on all preceding syllables.

Z&M advocated the long-distance spreading analysis in (a) (see their Association Rule (13b) for details); Zhang and Meng (2016), following Zhu (1995; 1996) take the (b) position, and view the situation in (3) as a case of contour displacement.

3.7 Summary and goals

In the section that follows, we present an analysis of Shanghainese tone melodies meant to unravel the puzzles presented above. We will postulate nothing about T1–T5 that distinguishes them, other than their underlying tonal representations. No reference to the phonemic characteristics of their syllable margins or the metrical properties of the expressions they occur in will be made in our analysis.

The account combines input-output correspondence requirements with markedness constraints on the shape of surface tone melodies. The distinct behavior of the five tone classes will follow from differences in their underlying tonal specifications, but most constraints will refer only to surface melodies themselves.

The phonological grammar we propose will provide answers to eight major questions that we elucidated above, which we call ‘QM’ and ‘QT1–T5’, and itemize below.

Questions to answer:

QM: Why does...

- a. every melody that exceeds two syllables and starts with T1–T4 end with the same *M–ML* sequence?
- b. T5 not do that?

QT1: Why does T1...

- a. not shift its right-edge tonal target tone to σ_2 of its tone melodies?
- b. have a final *ML* in its disyllable?

QT4: Why does T4...

- a. not retain its initial left-edge tonal target in polysyllables?
- b. produce the same outputs as T2 despite having only a single underlying /H/ tone (compared to the underlying /MH/ of T2)?

QT5: Why does T5...

- a. not shift its right-edge tonal target tone to σ_2 of its tone melodies?
- b. shift its final M target (or its entire *LM* contour) to the right edge of its polysyllables?

4 An OT Analysis

4.1 Introduction: tone shifting and faithfulness

This section endeavors to give an account of the tone melodies of Shanghainese that will address each of the questions that we left off with in the last section. Our analysis is framed in Optimality Theory (OT) (Prince and Smolensky, 1993; McCarthy and Prince, 1993a,b), which we will assume the reader is familiar with. Most of our questions pertain to the several deviations from the ‘straightforward’ behavior of T2 and T3, summarized below.

Remember that QT1a and QT5a were related to the two supposedly aberrant cases of T1 and T5; both answer ‘no’ to the second question in the table because they fail to shift their final

Tone	UR	disyllable	retains initial tone on σ_1 ?	shifts final tone to σ_2 ?	... $M-ML$ tetrasyllable?
T1	/HL/	$H-ML$	yes	no	yes
T2	/MH/	$M-H$	yes	yes	yes
T3	/LH/	$L-H$	yes	yes	yes
T4	/H/	$M-H$	no	yes	yes
T5	/LM/	$L-LM$	yes	no	no

tones to σ_2 in the way T2–T4 do. We begin our answer by pointing out a simple fact that holds, given the assumptions we have started with:

Fact 1: *Tones that end with H-tones shift. Tones that don't, don't.*

To be more precise: the underlying right-edge tonal target t of an initial syllable, σ_1 , is mapped onto the second syllable, σ_2 , of a tone melody if and only if t is a High tone. Since T1 and T5 do not end in an underlying H, this requirement does not apply to them. This observation alone will provide answers to QT1a and QT5a.

In addition to a constraint that demands that initial syllables realize final H-tones on the syllable to their right, we posit a constraint related to the retention of the initial tone target:

- (4) **ANCHOR-L(t)**: The leftmost input tonal target t must have a correspondent at the left edge of the output tone melody.

SHIFT-R(H): If H is the rightmost tonal target associated with σ_1 in the input, its output correspondent must be realized as the sole tonal target of σ_2 .

The constraints in (4) codify the two mapping relations exhibited most transparently by T2 and T3. For an input melody $/\sigma_1^{XH} + \sigma_2/$, **ANCHOR-L(t)** favors realization of the left-edge target X on σ_1 , while **SHIFT-R(H)** requires σ_2 to surface solely as H , yielding the familiar $X-H$ pattern. For the time being, these are the only constraints in our analysis of Shanghainese polysyllables that will refer to the identity or position of particular input tonal targets. The rest of the constraints introduced in this section evaluate the melodies themselves, regardless of the specific characteristics of whichever tone initiates them.

4.2 Constraining tone melodies (part 1): disyllabic mapping

Considering all the patterns exemplified above in Tables 4–7 and summarized in Table 3, two generalizations can be made about Shanghainese tone melodies across the board. First, for melodies longer than two syllables, each non-final syllable is ‘flat’ (i.e., it contains only one tonal target, as opposed to a contour syllable with more than one). Second, final syllables are *not* flat.

- (5) ***CONTOUR**: A syllable may not contain two distinct tonal targets.

FINALCONTOUR: The final syllable of a prosodic domain must contain two distinct tonal targets.

The conflict between these two constraints is not as theoretically contradictory as might seem; see, e.g., Zhang (2000) and Zhang (2004, p. 929) and references therein, who note a wide

variety of languages with a preference for contour tones on prosodic-final syllables. Shanghainese is clearly an example of such a language. Since the final syllable of longer tone melodies always falls (like T1–T4) or rises (like T5), FINALCONTOUR must dominate *CONTOUR. There is, however, an independently motivated exception to the general prohibition against flat final syllables: when SHIFT-R(*H*) requires a σ_2 to surface as *H* and σ_2 is also word-final (because the domain is disyllabic), flat final syllables are tolerated. (Recall, e.g., the T3-initial /n^{ji}s^{LH}na^{LH}/ (‘milk’), whose surface melody is *L–H*.) It follows that SHIFT-R(*H*) must dominate FINALCONTOUR. We have:

$$(6) \quad \text{SHIFT-R}(H) \gg \text{FINALCONTOUR} \gg * \text{CONTOUR}$$

Tables 8–10 illustrate the predictions these constraints make for disyllables. The ranking of ANCHOR-L(*t*) relative to the other constraints is not yet determined, but any candidate that violates that constraint is harmonically bounded by an otherwise equivalent candidate that satisfies it. For T2 and T3, the grammar generates the correct disyllables.

T3: / $\sigma_1^{LH} + \sigma_2$ /	SHIFT-R(<i>H</i>)	FINCON	*CONT	ANCH-L(<i>t</i>)
a. <i>LH–HL</i>	*!		**	
b. <i>LH–H</i>		*	*!	
☞ c. <i>L–H</i>		*		
d. <i>H–H</i>		*		*!

Table 8: Evaluation of selected T3-initial disyllables (with correct predictions).

In Table 8, candidate (a) satisfies FINALCONTOUR at the fatal expense of failing to shift its H-tone to σ_2 . Among candidates satisfying SHIFT-R(*H*), *CONTOUR favors (c) over (b). The candidate (d) is harmonically bounded, as it violates the same constraints as (c) and additionally violates ANCHOR-L(*t*). The attested *L–H* output is therefore accurately predicted as optimal. The reader may check that this analysis works just as well for T2-initial disyllables that start with M rather than L.

This accuracy does not extend to the disyllables of the other three tone classes. Consider the form of a T4 disyllable like /fa^ɿtsø^{MH}/ (‘development’), shown in Table 9. The grammar incorrectly predicts *H–H* to be the optimal surface form, since it satisfies both the faithfulness constraint and the shifting constraint. There is no explanation yet for why the first syllable of a T4 disyllable gets an M-tone, and thus currently no answer to QT4a.

T4: / $\sigma_1^H + \sigma_2$ /	SHIFT-R(<i>H</i>)	FINCON	*CONT	ANCH-L(<i>t</i>)
a. <i>H–LM</i>	*!		*	
☛ b. <i>H–H</i>		*		
☺ c. <i>M–H</i>		*		*!

Table 9: Evaluation of selected T4-initial disyllables (with incorrect predictions).

More generally, the constraints do not make accurate predictions for the two tone classes that do not trigger SHIFT-R(*H*), because they leave the final contours of T1 and T5 underdetermined. Consider the evaluation of a T1-initial disyllable like /t^hoŋ^{HL}çio^{HL}/ (‘all night’).

The result in Table 10 is only partly correct. Since T1’s underlying form /HL/ does not end in H, SHIFT-R(*H*) is moot, and the attested candidate (c) and the unattested (d–h) each incur

T1: $/\sigma_1^{HL} + \sigma_2/$	SHIFT-R(H)	FINCON	*CONT	ANCH-L(t)
a. $HL-HL$			*!*	
b. $HL-H$		*!	*	
 c. $H-ML$			*	
 d. $H-LM$			*	
 e. $H-MH$			*	
 f. $H-HM$			*	
 g. $H-LH$			*	
 h. $H-HL$			*	

Table 10: Evaluation of selected T1-initial disyllables (with incorrect predictions).

only one violation of the same constraint. Gross overgeneration occurs because the grammar cannot distinguish those candidates. The reader may check that the same problem would be encountered if the input were a T5 disyllable with $/\sigma_1^{LM} + \sigma_2/$. This means that QT1b and QT5b have yet to be answered.

We have made progress by accounting for the basic association pattern and answering two of the eight questions we set out to address, QT1a and QT5a. The other six questions remain open and we turn to these in the next subsection.

4.3 Constraining tone melodies (part 2): disyllables and beyond

We begin here with a discussion of QM: ‘Why does every melody that exceeds two syllables end with the same $M-ML$ sequence, except T5?’ As we will see, our answer to that question will also entail a partial answer to QT4. It will additionally show how both QT1b and QT5b turn out to be non-issues that pose no puzzle at all for a proper analysis of Shanghaiese tone melodies. The answer to QM can be found with the recognition of two additional basic facts about the sandhi patterns of Shanghaiese:

Fact 2: *Tone melodies don’t change direction unless they have to.*

Fact 3: *The only time they have to is when they need to avoid consecutive high-tones.*

Our earlier discussion did not touch much on QM, since the constraints proposed therein placed few restrictions on the identity of syllables that occur after the initial syllable-pair of a tone melody. E.g., for an input $/XH + \sigma_2 + \sigma_3 + \sigma_4/$, the constraints ANCHOR-L(t) and SHIFT-R(H) determine the realization of the first two syllables, but say nothing about the identity of the other two. Our constraint FINALCONTOUR rules out three possibilities for the final syllable, but that will still not explain why the attested terminal syllable ML is preferred over any of the other five possible contour-final sequences: LM , LH , HL , HM , or MH . Likewise, σ_3 could surface as L , H , or M , according to the present analysis, since none of these offends *CONTOUR; but only the last of these possibilities is the attested one. To appreciate why the $M-ML$ sequence is the only optimal way to conclude the melody, it is illuminating to consider the general shapes of the complete tetrasyllabic melodies across the five tone classes, shown in Figure 2.

The melodies in Figure 2 reveal a strong preference for consistent melodic directionality. T1 descends throughout the domain, while T5a does the opposite. T2, T3, T4, and T5b rise to a High peak, then immediately begin to descend. What is clearly being systematically avoided is any reversal in melodic direction without ‘justification.’ And there is only one kind of justification.

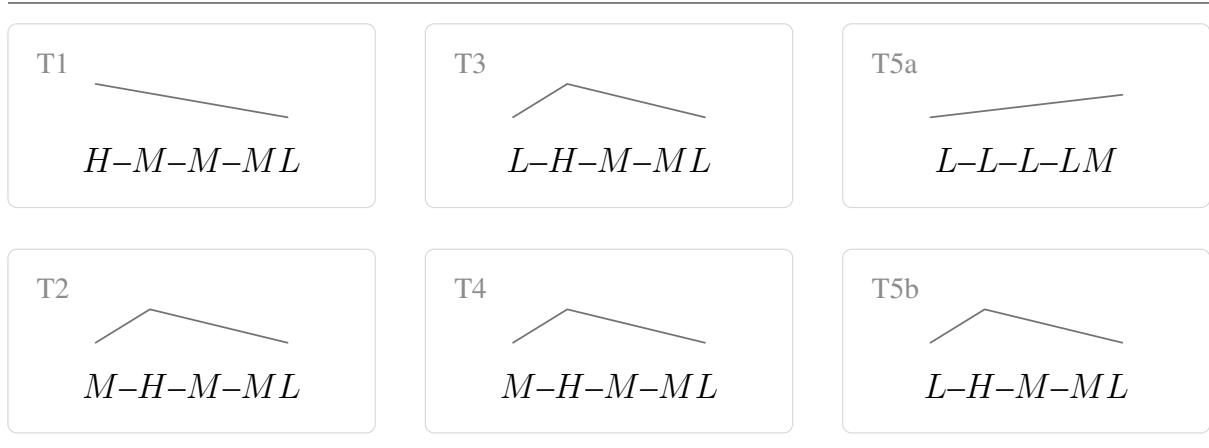


Figure 2: Directionality in the tetrasyllabic tone melodies. Melodies change direction if and only if they reach a High-tone peak.

(7) $OCP(H)$: High tonal targets may not be adjacent.

*MELODYREVERSAL: Do not reverse the direction of a tonal melody.

The ranking between these constraints can be deduced from the melodies in Figure 2. Once an H peak has been reached, continuing on with a consecutive H-tone would violate $OCP(H)$. The melody instead turns downward, incurring a violation of *MELODYREVERSAL. The fact that the reversal is preferred establishes:

(8) $OCP(H) \gg *MELODYREVERSAL$

A second ranking relation follows from the typical patterns attested for T2 and T3. For a T3-initial input (UR: /LH/), an unattested candidate such as *L-L-LM* would avoid any melodic reversal and would retain its initial *L*, but would fail to realize its H-tone on σ_2 . The fact that T3 melodies do not exhibit such a pattern allows us to infer:

(9) $SHIFT-R(H) \gg *MELODYREVERSAL$

Together with the rankings we established before, a partial order at this stage is shown in the Hasse diagram in Figure 3. The ranking of $ANCHOR-L(t)$ is undetermined at this point. No ranking relation between $FINALCONTOUR$ and *MELODYREVERSAL has been established either, nor can we determine any ranking between $OCP(H)$ and $SHIFT-R(H)$. (In the tableaux in Tables 11 and 12, $ANCHOR-L(t)$ and candidates that violate it are omitted because they are harmonically bounded. This should not be interpreted as a claim that $ANCHOR-L(t)$ is undominated. Its ranking becomes relevant in the analysis of T4 below.)

These constraints correctly derive the output *L-H-M-ML* for a T3 trisyllable in the following way: (i) $ANCHOR-L(t)$ fixes the first syllable as *L*; (ii) $SHIFT-R(H)$ fixes the second syllable as *H*; (iii) $OCP(H)$ prevents the occurrence of an additional High tone immediately after that; (iv) $FINALCONTOUR$ prohibits ‘flat’ final syllables; and (v) *MELODYREVERSAL selects *ML* as the terminal syllable because any other contour syllable would force an additional reversal of the tone melody.

The results for T2 (UR: /MH/), are analogous to those for T3 in Table 11. The same constraint interactions extend from trisyllabic to tetrasyllabic domains, as illustrated in Table 12.

Recall QM: ‘Why do T1–T4-initial trisyllables and tetrasyllables all end in *(M-)ML* and why doesn’t T5 do that?’ There is a short answer and a longer one. The short answer is: because T1–T4 all have high tones, and T5 does not.

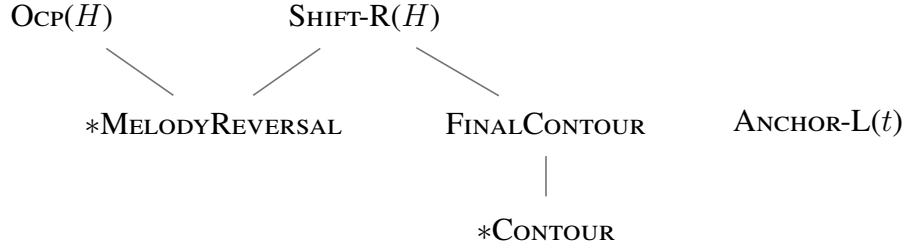


Figure 3: Ranking relations established after (9), (8), and (6). The ranking of $\text{ANCHOR-L}(t)$ is not yet determined.

T3: $/\sigma_1^{LH} + \sigma_2 + \sigma_3/$	OCP(H)	SHIFT-R(H)	FINCON	*CONT	*MELREV
<i>L-L-LM</i>		*!		*	
<i>L-L-MH</i>		*!		*	
<i>L-H-L</i>			*!		*
<i>L-H-LM</i>				*	**!
<i>L-H-ML</i>				*	*
<i>L-H-M</i>			*!		*
<i>L-H-MH</i>				*	**!
<i>L-H-HL</i>	*!			*	*
<i>L-H-HM</i>	*!			*	*
<i>L-H-H</i>	*!		*		

Table 11: Evaluation of T3-initial trisyllables. Candidates violating $\text{ANCHOR-L}(t)$ are harmonically bounded and are omitted.

The longer answer is: the facts that inspired QM, including the discrepancy between the behavior of T1–T4 versus T5 are due to the conjoint pressure of (a) the avoidance of consecutive High tones, (b) the avoidance of melody reversal, and (c) the avoidance of flat syllables occurring finally. A tetrasyllable melody whose dominant syllable contains an H-tone will have an H-tone somewhere in it. After the High tone, (a–c) leave only one way to finish the melody: it must not remain high (OCP); it cannot go down to L immediately with no place to go except up (since this would force a melody reversal); and it cannot go down to M without falling further (otherwise it would need to end with a flat syllable). This logic not only gives us the answer to QM, but also gets us the answer to QT1b. We turn to this forthwith.

4.4 Analyzing T1

We can now improve our earlier account of T1. The behavior of T1 originally seemed puzzling because it doesn't simply shift its right-edge tone to the σ_2 of its disyllable like T2–T4 do; instead, it immediately follows the σ_1 with the *ML* terminal sequence that T2–T4 reserve for longer melodies. QT1 was: why doesn't T1 shift and why does it terminate specifically with the sequence *ML*? We answered QT1a right away: T1 doesn't shift because it doesn't end with an H like T2–T4 do. As for QT1b, the final L in the T1 disyllable *H-ML* is not there due to shifting or faithfulness; it's due to the conspiracy of factors (a–c), discussed immediately above: since the T1-initial disyllable *begins* with a High tone, it reaches its peak immediately. There is nowhere to go but down, and only one way to finish the melody without reversing it or ending it without a contour. (See Table 13.) We are thus able to predict the behavior of the T1 disyllable

T2: $/\sigma_1^{MH} + \sigma_2 + \sigma_3 + \sigma_4/$	OCP(<i>H</i>)	SHIFT-R(<i>H</i>)	FINCON	*CONT	*MELREV
<i>M-M-M-ML</i>		*!		*	
<i>M-H-L-L</i>			*!		*
<i>M-H-L-LM</i>				*	**!
<i>M-H-M-L</i>			*!		*
<i>M-H-M-LH</i>				*	**!
☞ <i>M-H-M-ML</i>				*	*
<i>M-H-M-M</i>			*!		*
<i>M-H-H-L</i>	*!		*		*
<i>M-H-H-ML</i>	*!			*	*

Table 12: Evaluation of T2-initial tetrasyllables. Candidates violating ANCHOR-L(*t*) are harmonically bounded and are omitted.

with the exact same logic that we used to predict T2–T4 without introducing any “additional machinery” of the sort mentioned by Zhang (2007) in the previous section.

T1: $/\sigma_1^{HL} + \sigma_2/$	OCP(<i>H</i>)	SHIFT-R(<i>H</i>)	FINCON	*CONT	*MELREV
<i>HL-ML</i>				**!	
<i>H-LM</i>				*	*!
☞ <i>H-ML</i>				*	
<i>H-M</i>			*!		
<i>H-MH</i>				*	*!
<i>H-HM</i>	*!			*	

Table 13: Evaluation of selected T1-initial disyllables. Candidates violating ANCHOR-L(*t*) are harmonically bounded omitted.

4.5 Analyzing T4

Our answers to QM and T1a relied on the observation that Shanghainese tone melodies will do whatever it takes to avoid placing H-tones next to one another. In the cases of T2 and T3, that meant reversing the direction of the tonal melody when the domain was so large that there was no other way to avoid offending OCP(*H*). The disyllable initiated by T4 not only provides more evidence of how adamantly this language prefers to avoid such an offense but also shows us which avoidance strategy it prefers when it has a choice.

Recall Q4: Why does T4 not keep its left-edge tonal target in polysyllables? Why do its disyllable, trisyllable, and tetrasyllable melodies all start with an M, just like those of T2, even though T4 and T2 have different underlying tonemes? We assume that T4 has a single underlying tonal target $/H/$. For an input $/\sigma_1^H + \sigma_2/$, SHIFT-R(*H*) requires σ_2 to surface as *H*, while ANCHOR-L(*t*) favors an output whose left edge remains *H*. Among the candidates satisfying both requirements are *H-H*, *HM-H*, and *HL-H*. The first is excluded by OCP(*H*). The latter two avoid an OCP violation, but each introduces a reversal in melodic direction: both melodies fall from the initial H, then rise again to the H in the second syllable. In addition, they both commit an extra violation of *CONTOUR, since the epenthetic L or M will give σ_1 a falling tone.

The attested $M-H$ avoids both the contour and the melodic reversal, but only by sacrificing a faithful realization of its left edge.

The behavior of the T4 disyllable therefore forces us to conclude that either:

- (10) a. $*\text{MELODYREVERSAL} \gg \text{ANCHOR-L}(t)$, or
 b. $*\text{CONTOUR} \gg \text{ANCHOR-L}(t)$, or
 c. both

For now, we will choose (a) and operate under the assumption that our constraints have the rankings shown in Figure 4. We will return to a discussion of other possible grammars below in § 6.3.

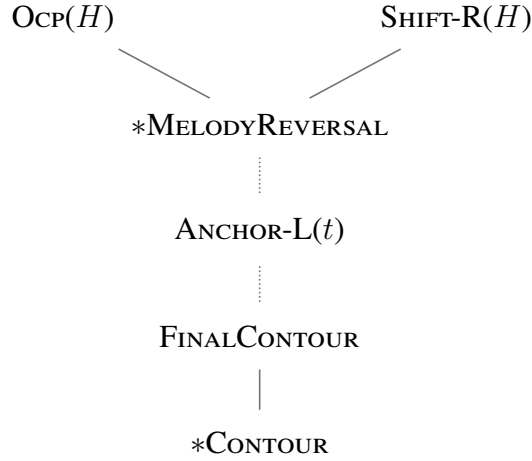


Figure 4: Updated constraint hierarchy. Dotted lines indicate rankings tentatively assumed.

The grammar in place is still incomplete, since while it would determine why the first half of a T4 disyllable cannot remain High, it does not say why it can't be Low. For an underlying form $/\sigma_1^H + \sigma_2/$, the competing outputs $M-H$ and $L-H$ both satisfy and violate the same constraints. We claim that the distinction between them will follow from two general register-related markedness constraints that will formalize what we suggested in the previous section: M is a default tone.

- (11) $*\text{HIGH}$: Assign one violation for every High tonal target.

$*\text{LOW}$: Assign one violation for every Low tonal target.

Because M violates neither constraint, the two constraints jointly make Mid the unmarked tonal register. Their relative ranking is not relevant to the evaluation of T4 disyllables: $M-H$ and $L-H$ contain the same number of High targets, while the latter incurs one additional violation of $*\text{LOW}$. (See Table 14.)

It goes almost without saying that the aforementioned pair of constraints cannot outrank the only faithfulness constraint in our analysis. If it did, then no word of Shanghainese would begin with an H- or L-tone.

- (12) $\text{ANCHOR-L}(t) \gg * \text{HIGH}, * \text{LOW}$

We can further deduce that the aversion to H and L-tones cannot possibly be stronger than the proclivity for word-final contours. If it were so, then citation tones like MH (T2) and LM (T5) would never surface, not to mention the ML contour with which most Shanghainese tone melodies conclude.

(13)

FINALCONTOUR $\gg$ *HIGH, *LOW

T4: $ \sigma_1^H + \sigma_2 $	OCP(H)	SHIFT-R(H)	*MELREV	ANCH-L(t)	*HIGH	*LOW
<i>H-H</i>	*!				**	
<i>H-M</i>		*!			*	
<i>H-MH</i>		*!	*		**	
<i>HM-H</i>			*!		**	
<i>HL-H</i>			*!		**	*
☞ <i>M-H</i>				*	*	
<i>L-H</i>				*	*	*!

Table 14: Evaluation of selected candidates for T4-initial disyllable. (Since all candidates that satisfy FINALCONTOUR will fatally violate SHIFT-R(H), the constraints FINALCONTOUR and *CONTOUR are omitted here.)

This analysis provides us with all we need to give a complete answer to QT4: T4 surfaces as the same disyllable as T2 due to the combination of pressure to (a) avoid adjacent High tones, (b) avoid a reversal of the tonal melody, and (c) prefer Mid tones over Low or High tones when faced with the necessity to populate a syllable with tonal material.⁴ The longer T4 melodies follow from the same constraints already discussed for T2, since they are identical. (See Table 12 above and the related discussion.)

4.6 Analyzing T5

The behavior of T5 differs significantly and fundamentally from that of T1–T4. Based on what we have already said above, we can identify one basic reason for this: T5 has no High tone. We observed how this furnished an answer to QT5a: just like T1, T5 does not participate in the rightward tone-shifting that T2–T4 do because it has no right-edge H-tone to shift. But T1 still has an H-tone on its left-edge, which is why the syllables that appear after its pitch-peak look exactly like those that follow the H-peak in T2–T4. T5 has no H at all and can satisfy OCP(H) quite easily without reversing the direction of its melodies. A few questions do remain that can be answered with specific rankings of the constraints we already have.

Based on the disyllabic form for T5, we can note an additional ranking relation. A candidate such as *L-MH* preserves the left-edge L and maintains a consistently rising melody, while containing fewer Low targets than the actual T5 disyllable, *L-LM*. Its only disadvantage is that it contains an extra H target instead of an extra L. Thus we know that:

(14)

*HIGH $\gg$ *LOW.

This conclusion will leave us with the complete grammar shown in Figure 5.

Recall QT5b: Why does T5 shift (all or part of) its underlying *LM* syllable to the end of each of its tone melodies? The answer is: it doesn't. This is again a coincidental similarity that falls out of the preferences regarding the directionality of tone melodies and the relative markedness of the tone targets themselves. T5 does not shift anything. It simply stays low and flat until its final syllable is compelled to surface as some kind of contour. Where that need arises, the

⁴We are assuming here that toneless syllables are not an option, echoing the Well-Formedness Condition of Zee and Maddieson (1979), mentioned earlier. This could be expressed as a constraint, though we forgo formalizing that detail here.

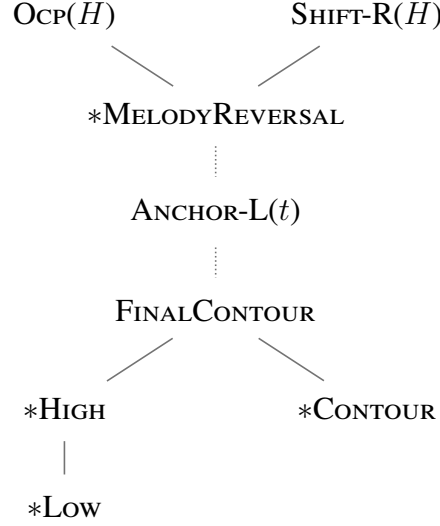


Figure 5: Final constraint hierarchy. Solid lines indicate rankings deduced from the evidence and dotted lines indicate rankings tentatively assumed.

‘cheapest’ contours in light of the relative economy scale $M > L > H$ are ML and LM . But choosing ML would mean reversing the tonal melody, which is exactly why LM is the only surface form for σ_2 of the disyllable. (See Table 15)

T5: $/\sigma_1^{LM} + \sigma_2/$	*MELREV	ANCH-L(t)	FINCON	*CONT	*HIGH	*LOW
$L-L$			*!			**
$L-M$			*!			*
$L-LM$				*		**
$L-LH$				*	*!	**
$L-ML$	*!			*		**
$L-MH$				*	*!	*
$M-L$		*!	*			*
$M-LM$	*!	*		*		*
$M-ML$		*!		*		*

Table 15: Evaluation of selected candidates for T5-initial disyllables.

The same logic gives empirical coverage to longer T5 domains. ANCHOR-L(t) preserves the L-tone, *HIGH prevents the melody from rising to a High target, FINALCONTOUR requires a contour at the right edge, and *MELODYREVERSAL favors a melody whose direction remains consistent. The resulting tetrasyllabic output is therefore just $L-L-L-LM$. The surface configuration that gives the appearance of displacement (Zhu, 1995; Zhang and Meng, 2016) or long-distance spreading (Zee and Maddieson, 1979) instead emerges from independently motivated general restrictions on faithfulness, markedness, and melodic directionality. (See Table 16.)

4.7 A synopsis of T1–T5 tone melodies

We ended § 3 with an itemized list of questions, which we now revisit.

QM: Why does...

T5: $/\sigma_1^{LM} + \sigma_2 + \sigma_3 + \sigma_4/$	*MELREV	ANCH-L(<i>t</i>)	FINCON	*CONT	*HIGH	*Low
<i>M-M-M-M</i>		*!	*			
<i>M-M-M-ML</i>		*!		*		*
<i>L-M-M-ML</i>	*!			*		**
<i>L-L-L-LM</i>				*		****
<i>L-M-M-MH</i>				*	*!	*
<i>L-L-L-LH</i>				*	*!	****
<i>L-L-L-ML</i>	*!			*		****
<i>L-L-L-M</i>			*!			* * *
<i>L-L-L-MH</i>				*	*!	* * *

Table 16: Evaluation of selected candidates for T5-initial tetrasyllables.

- a. every melody that exceeds two syllables and starts with T1–T4 end with the same *M–ML* sequence?
- b. T5 not do that?

The *M–ML* sequence occurs at the end of all T1–T4 tone trisyllables and tetrasyllables because that is the only way to end the melody without either (a) reversing the falling direction of the melody and/or (b) ending the melody with a syllable that has no contour at all. The reason the longer T1–T4 melodies must end with a falling trajectory is that they have already risen to the top of the register, and, due to a ban on adjacent High tones, they cannot remain there. The reason T1–T4 all rose to the top of the register in the first place is that they all have lexical High tones. T5 is the exception exactly because it is the only one that does not have a High tone. Ergo, it has no need to reach the top of the register, nor any need to start falling. Instead, it may remain Low until the end of its melody, where the only thing demanded of it is some contour on the final syllable, which will, of course, be a rising one. Naturally, the most economical rise from L is to the closest destination target, namely M.

QT1: Why does T1...

- a. not map its right-edge tonal target tone to σ_2 of its tone melodies?
- b. have a final *ML* in its disyllable?

QT5: Why does T5...

- a. not shift its right-edge tonal target tone to σ_2 of its tone melodies?
- b. shift its final M target (or its entire *LM* contour) to the right edge of its polysyllables?

The short answers to QT1a and QT5a are the same: because they don't end in High tones. The oft-discussed 'spreading' or 'shifting' process in Shanghainese only maps H-tones from the right edge of an initial syllable to the next syllable. There are no High tones on the right edge of T1 or T5, so the alternation does not affect them.

As for QT1b, according to the account we are advocating, the remark from Zhang that we referenced in § 3.5 about "the L from the initial syllable and the boundary L [having] merged into one" is half correct and half incorrect. As we hinted in the title of that section, we agree that the L in the T1 disyllable *M–ML* is indeed a boundary tone, but it has not merged with anything. A T1 disyllable might look like it is shifting its right-edge tone in the usual way, except for an extra M that somehow got added to the picture, but the fact that the underlying form of T1 and the second half of a T1 disyllable both have a Low tone is mere coincidence. The *ML* tone occurs at the end of a T1 disyllable for the same reason it occurs at the end of most other lexical polysyllables in Shanghainese. (See our answer to QM above.)

The puzzle of QT5b is likewise more of an illusion than an actual theoretical complication; a T5 tone might appear to be ‘shifting’ its left-edge tonal target(s) so that it/they appear(s) domain-finally, but in fact it is simply doing nothing since it is under no obligation to shift its right-edge tonal target because that target is not High. The low rise at the end of T5 melodies shows up for reasons we mentioned in our answer to QM but will reiterate: the melody must end with some contour, there is nowhere to go but up, and a rise to Mid is less costly than a rise all the way to High.

QT4: Why does T4...

- a. not retain its initial left-edge tonal target in polysyllables?
- b. produce the same outputs as T2 despite having only a single underlying /H/ tone (compared to the underlying /MH/ of T2)?

Due to a restriction on adjacent High tones, a T4 tone melody can’t begin with *H–H*. Thus, a choice must be made as to which is more important: allowing the H tone to migrate to the second syllable or allowing it to stay with the first. Shanghainese prefers the former, and the initial syllable is instead occupied with a default tone, which is Mid. The remainder of the longer melodies follows the usual playbook, hence their resemblance to those of T2.

5 Citation forms and underlying tones

While the constraint hierarchy developed in § 4 was motivated primarily to account for the system of Shanghainese polysyllabic tone melodies, most of the citation forms of T1–T5 follow from that hierarchy given the underlying forms we have assumed without much modification.

The underlying representations of the five Shanghai tones have been the subject of considerable debate. We have mostly avoided this issue other than to state our own assumptions in § 2.2 and § 4.1, repeated in Table 17 for convenience.

Tone	Underlying form	Citation form
T1	/HL/	<i>HM</i>
T2	/MH/	<i>MH</i>
T3	/LH/	<i>LM</i>
T4	/H/	<i>H</i>
T5	/LM/	<i>LM</i>

Table 17: Underlying and derived citation-tone melodies, according to the present analysis.

We will return to the motivation for these assumptions and the controversy that surrounds them in the next section. First, we address the derivation of the surface forms from the tonemic forms. Starting with T5, Table 18 shows how its underlying form will surface faithfully without further assumptions.

Table 19 shows how the tones T1 and T3 do *not* surface faithfully in that they do not realize the full pitch excursion of their underlying melodies. Instead, the grammar selects *HM* for T1 and *LM* for T3. These mappings follow from markedness relations already independently proposed for the sandhi analysis above: in each case, a less-marked M tone is substituted for the final /L/ and /H/, respectively. The predicted outputs are consistent with traditional pitch-scale

T5: /LM/	ANCH-L(<i>t</i>)	FINCON	*CONT	*HIGH	*LOW
<i>H</i>	*!	*		*	
<i>M</i>	*!	*			
<i>L</i>		*!			*
☞ <i>LM</i>			*		*
<i>LH</i>			*	*!	*
<i>ML</i>	*!		*		*
<i>MH</i>	*!		*	*	
<i>HL</i>	*!		*	*	*
<i>HM</i>	*!		*	*	

Table 18: Monosyllabic outputs of citation forms for T5.

descriptions of the phonetic realizations, which include 53 or 52 for T1 and 113 or 223 for T3 (Xu and Tang, 1988, p. 8), as well as with the more recent descriptions of Ling (2025), who characterizes the surface realizations of T1 and T3 as *HM* and *LM*, respectively.⁵ (See Table 2 in § 2.)

T1: /HL/	FINCON	*HIGH	*LOW	T3: /LH/	FINCON	*HIGH	*LOW
<i>H</i>	*!	*		<i>L</i>	*!		*
<i>HL</i>		*	*!	☞ <i>LM</i>			*
☞ <i>HM</i>		*		<i>LH</i>		*!	*

Table 19: Monosyllabic outputs of the citation forms for T1 and T3.

There are, however, some incorrect predictions. Table 20 shows how, for T2, the constraint hierarchy established thus far selects *ML* rather than the faithful *MH* due to the ranking *HIGH ≫ *LOW. The grammar therefore incorrectly predicts a falling citation form for an underlyingly rising tone.

T2: /MH/	ANCH-L(<i>t</i>)	FINCON	*CONT	*HIGH	*LOW
<i>LM</i>	*!		*		*
<i>M(M)</i>		*!			
☛ <i>ML</i>			*		*
☺ <i>MH</i>			*	*!	

Table 20: Incorrect output for citation form of T2.

A general target-identity constraint would not solve this problem, since any constraint that could preserve the final H of T2 /MH/ would certainly do the same for the underlying final target tones of T1 and T3, which, as we have just seen, were realized as M in a way that we were

⁵We are treating these alternations as genuine phonological reduction rather than as merely phonetic reduction. While this is not the only possible interpretation, the reduced endpoint corresponds to the same M element independently required elsewhere in the sandhi system and we view the recurrence of M as a reduced target across otherwise opposite underlying contours as favoring a single representational generalization rather than seeing it as two unrelated cases of consistent phonetic undershoot.

pleased with because it comports with the general phonological reduction that is observed in acoustic examinations of those tone classes. But while substituting M for H or L in a rising or falling tone is phonological reduction, substituting L for H is not. On the other hand, if /MH/ reduced its final target tone and was realized as *MM*, it would cease to be a contour and would just be a flat Mid tone. What must remain faithful, then, is not the exact value of every tonal target, but the direction or ‘Contour’ of the tonal melody, in the sense of the formal parameters advocated by Rose (1990) and Zhu (1995). We can capture this:

- (15) ID(Contour) : The direction of the tonal contour of the input must be preserved in the output; an output may be α -contour ($\pm$ rising, $\pm$ falling) only if its corresponding input is α -contour.

The constraint ID(Contour) allows the magnitude of a pitch excursion to change without allowing the direction of that excursion to reverse.⁶ The required ranking is:

- (16) $\text{ID(Contour)} \gg *_{\text{HIGH}} \gg *_{\text{Low}}$

Thus, the results we welcomed for T1 and T3 are preserved because the inputs and optimal outputs for each will be either falling contours (like T1) or rising (like T3). By contrast, realizing /MH/ as *ML* violates the constraint because a rising input is realized as a falling output. Note that this constraint will also prevent T4 /H/ from surfacing as a contour like *HM*, since such realization would satisfy FINALCONTOUR , but would violate ID(Contour) because a non-falling input is realized as a falling contour. We thus know that:

- (17) $\text{ID(Contour)} \gg \text{FINALCONTOUR}$

Table 21 illustrates how this last additional constraint preserves the desirable predictions regarding the input-output mismatches of T1 and T3, while ensuring the faithfulness of T2 and T4.

T1: /HL/	ID(Contour)	$\text{ANCH-L}(t)$	FINCON	* CONT	* HIGH	* Low
<i>HL</i>				*	*	*!
☞ <i>HM</i>				*	*	
T2: /MH/	ID(Contour)	$\text{ANCH-L}(t)$	FINCON	* CONT	* HIGH	* Low
<i>ML</i>	*!			*		*
☞ <i>MH</i>				*	*	
T3: /LH/	ID(Contour)	$\text{ANCH-L}(t)$	FINCON	* CONT	* HIGH	* Low
☞ <i>LM</i>				*		*
<i>LH</i>				*	*!	*
T4: /H/	ID(Contour)	$\text{ANCH-L}(t)$	FINCON	* CONT	* HIGH	* Low
☞ <i>H</i>			*		*	
<i>M</i>		*!	*			
<i>HM</i>	*!			*	*	

Table 21: Citation forms for T1–T4, illustrating the effect of ID(Contour) .

⁶Note that despite some similarities, ID(Contour) and * MELODYREVERSAL are quite distinct. The latter is an output markedness constraint that is oblivious to underlying forms whereas the former evaluates correspondence between the input and output.

This single amendment to the grammar we proposed in the previous section allows the same basic constraint system that derives the tone melodies to account for monosyllabic citation tones of tone classes T1–T5, including differences between the proposed underlying representations and the surface forms.

A final formal detail concerns how $\text{ID}(\text{Contour})$ would evaluate polysyllabic sandhi domains. In particular, it is necessary to be specific about whether $\text{ID}(\text{Contour})$ is a local faithfulness constraint evaluating the features of a single initial syllable or one that evaluates trajectories of an entire tone domain. We choose the latter and interpret $\text{ID}(\text{Contour})$ as evaluating the entire sandhi domain; it disregards syllable boundaries and penalizes melodies whose contour is not the same as the contour of the initial input. This follows the spirit of our analysis above as well as the autosegmental tradition of Yip (1980, 2002) and Chen (2000) in analyzing Shanghainese left-dominant sandhi as a domain-level tone-mapping process where the contour behaves as a melodic unit.

While most longer melodies will violate $\text{ID}(\text{Contour})$, this will not preclude desirable predictions as long as we assume that the constraint is outranked by the markedness and correspondence constraints that affect the tone association pattern we discussed at length in § 4.1–4.3.

$$(18) \quad \text{OCP}(H), \text{SHIFT-R}(H) \gg \text{ID}(\text{Contour})$$

In polysyllabic tone melodies, the pressure to shift a High target rightward will override contour faithfulness. In monosyllabic citation forms, where $\text{SHIFT-R}(H)$ and $\text{OCP}(H)$ do not play a role, $\text{ID}(\text{Contour})$ is decisive, preserving the underlying contour of the input.

Note that our decision to interpret this faithfulness constraint as a global constraint rather than a local one is not made merely for continuity purposes or out of theoretical allegiance; it has immediate empirical ramifications. E.g., suppose that $\text{ID}(\text{Contour})$ only checked if the contour of the initial syllable in the output matched the contour of the initial, dominant syllable of the input. The undesired predictions for a T3 disyllable are shown in Table 22; cf. the predictions in Table 23 under the global interpretation of the same constraint.

T3: $/\sigma_1^{LH} + \sigma_2/$	$\text{SHIFT-R}(H)$	$\text{ID}(\text{Contour})_{[\text{loc}]}$	*MELREV	ANCH-L(t)	FINCON	*CONT	*HIGH	*LOW
$L-L$	*!	*			*			**
$L-LH$	*!	*				*	*	**
☛ $LM-H$					*	*	*	*
$L-HL$	*!	*	*			*	*	**
$L-HM$	*!	*	*			*	*	*
☺ $L-H$		*!			*		*	*
$M-H$		*!		*	*		*	

Table 22: Evaluation of selected T3-initial disyllables with local interpretation of $\text{ID}(\text{Contour})$ (demonstrating incorrect predictions).

This grammar constitutes the basic analysis of the five tone classes and the melodies they initiate. One outstanding issue—the alternative T5 melody we called ‘T5b’ above—still requires explanation. In the section that follows, we will address that point. We will also elaborate on some of the assumptions that our account relied upon, as well as some alternatives to our assumptions about underlying forms, constraints, and their rankings that we proposed above.

T3: $/\sigma_1^{LH} + \sigma_2/$	SHIFT-R(<i>H</i>)	ID(Contour)	*MELREV	ANCH-L(<i>t</i>)	FINCON	*CONT	*HIGH	*LOW
<i>L-L</i>	*!	*			*			**
<i>L-LH</i>	*!					*	*	**
<i>LM-H</i>					*	*!	*	*
<i>L-HL</i>	*!	*	*			*	*	**
<i>L-HM</i>	*!	*	*			*	*	*
$\mathbb{E}$ <i>L-H</i>					*		*	*
<i>M-H</i>				*!	*		*	

Table 23: Evaluation of selected T3-initial disyllables with domain-level ID(Contour) (demonstrating correct predictions).

6 Discussion

6.1 Assumptions of the present analysis

Traditional Sinitic dialectological descriptions of tone and tone sandhi like the ones we mentioned in § 2 take citation tones at face value, whereas generative analyses have typically sought to reduce the tonal inventory or derive parts of it from a smaller set of tonal primitives. Proposals regarding the underlying tones of Shanghainese especially differ over the representations of T2, T4, and T5. Table 24 compares the assumptions we began our analysis with to those in other analyses.

	T1	T2	T3	T4	T5
Present analysis	/HL/	/MH/	/LH/	/H/	/LM/
Z&M (1979)	/HL/	/MH/	/LH/	/H/	/LH/
Lu (1987)	/H/	/L/	/L/	/L/	/L/
S&S (1990)	/HL/	/MH/	/LH/	/MH/	/LH/
Duanmu (1997)	/HL/	/LH/	/LH/	/LH/	/LH/
Duanmu (1999)	/H/	/L/	/L/	/L/	/L/

Table 24: Proposed underlying representations of tone classes T1–T5.

Duanmu (1997) assumed /HL/ and /LH/ to be the only underlying forms and derived the rest of the distinctions from segmental properties and assigning a central role to metrical structure and stress. In later work, Duanmu (1999) derives those contours through a polarity constraint that supplies an opposing tonal target, yielding *HL* from /H/ and *LH* from /L/. Earlier work of Lu (1987) assumes those same URs and derives the surface differences through underspecification, default tonal specifications, and rules sensitive in part to segmental properties.

The present analysis deliberately makes the opposite sort of trade-off. Rather than minimizing the underlying tonal inventory and recovering the distinctions among the tone classes through segmental, metrical, or other representational properties, we retain more tonal information in the underlying representations themselves. This allows the differences among the sandhi patterns to be expressed, as far as possible, through independently motivated constraints on tonal correspondence and the shape of the resulting melodies.

Selkirk and Shen (1990), for their part, analyze T4 and T5 as the checked-syllable counterparts of T2 and T3, respectively, and therefore assign T4 the same underlying representation as

T2, and T5 the same representation as T3. As discussed in § 3.3, we do not adopt their analysis of T4. Instead, following Z&M, we take T4 to be underlyingly /H/. The principal difficulty raised by this assumption is the fact that T4-initial polysyllables are nevertheless identical to those initiated by T2. We argued in § 4.5 that this congruence need not reflect an underlying identity between the two tone classes, but instead follows from the interaction of independently motivated constraints in the grammar.

This leaves T5. All the generative analyses mentioned share the that view the underlyingly tonal properties of T5 are identical to those of T3 and derive their distinct surface properties from non-tonal information. Z&M and S&S, for example, both represent T5 and T3 as /LH/. We instead take T5 to be underlyingly /LM/.⁷

The postulate that T3 and T5 have distinct underlying representations does considerable work in our analysis; assuming that T2–T4 are the only tone classes that have underlying final H targets allows us to state our ‘Fact 1’ in § 4 and allows *SHIFT-R(H)* to operate on those three tone classes while not affecting T1 or T5. This in turn permits their sharply different sandhi behavior to follow from a difference in tonal representation rather than from an additional class-specific rule or from independently stipulated segmental or metrical properties. Because this aspect of the analysis is critical, we wish to briefly consider (a) the degree of independent support there is for the assumption that the final target tone of T5 is M, not H, and (b) the hypothetical consequences that our analysis would face if that assumption were not true.

6.2 Evidence concerning the representation of T5

The view that T3 and T5 are underlyingly distinct is not motivated solely by their difference in sandhi behavior. There are independent reasons to believe it. In this subsection, we discuss the motivation for our stance as well as a fallback analysis that is still compatible with the regimen of constraints proposed in § 4 for any reader who is still not convinced of a tonal distinction between T3 and T5.

6.2.1 Phonetic descriptions and acoustic evidence

Firstly, as we saw in Table 2 in § 2.1, traditional descriptions of T5 are almost always transcribed as low tone whose contour stays in the the low register, most commonly 12, rather than with a rise extending into the upper portion of the pitch range. In contrast, the final target of T3 is described as higher and transcriptions of T3 are usually 13 or 14.

Acoustic descriptions of T5, both in citation productions (Zee, 1988; Zhang and Meng, 2016) and in carrier-sentence contexts (Ling, 2025), are more ambiguous. All are perfectly compatible with a realization of T5 whose endpoint does not require an upper-register target, but generally the acoustic endpoints of T3 and T5 are measured at about the same F_0 height. However, as we discussed in § 5, the analysis we’ve laid out predicts the phonological reduction of T3 (/LH/ → LM), so we expect the final tonal targets in the citation forms of T3 and T5 to be the same.

⁷This assumption is not totally without precedent. Hsieh (2007, p. 83; 91–97), represents the checked rising T5 as underlyingly /LM/. To be sure, though, he assumes that T3 is /LM/ underlyingly, too, and his assumptions about tonemic qualities are more in line with traditional analyses’ faith in citation forms. Huang (2011), on the other hand, explicitly argues against the reductionist stance of Selkirk and Shen and assumes the same five underlying representations for T1–T5 that we have. He derives the surface patterns of T4 and T5 through an interaction of tonal representation with metrical structure and stress, rather than through tonal mapping constraints alone, as in our analysis, but it provides at least one independent precedent for the underlying T3–T5 distinction advocated in the present account.

For these reasons, we do not see the acoustic similarity of T5 and T3 as a theoretical problem and are inclined to believe that the traditional tonal distinction between T5 and T3 reflects an underlyingly real one.

6.2.2 Evidence from right-dominant sandhi

Our account in § 4 showed how the assumptions of the present analysis in Table 24 mesh well with a purely tonal account of Shanghainese left-dominant sandhi. Further evidence that T3 and T5 are tonemically distinct comes from their behavior under right-dominant sandhi. As we mentioned in § 2.2 there is a second type of sandhi in Shanghainese that operates not at the word level but at the phrase level. Phrase-level sandhi is right-dominant; the rightmost element of the domain retains its citation tone, while those to the left surface as sandhi tones. Phrase-level, right-dominant sandhi involves no complex melodies like those in word-level sandhi, and non-dominant tones do not completely lose their own tonal identity; each tone class simply has its own sandhi form. Some have claimed that this variety of sandhi is merely phonetic reduction (Zhang and Meng, 2016; Ling and Liang, 2018; Yan, 2018), but whatever the case may be, the right-dominant sandhi forms of T3 and T5 are not the same tone; T5's right-dominant sandhi form is a Low tone while T3 surfaces as a Mid tone. (See Table 25.)

input		output
T1 + X	$\rightarrow$	HX
T2 + X	$\rightarrow$	HX
T3 + X	$\rightarrow$	MX
T4 + X	$\rightarrow$	HX
T5 + X	$\rightarrow$	LX

Table 25: Shanghai disyllabic right-dominant sandhi (adapted from Xu and Tang 1988, p. 25).

This contrast cannot be attributed simply to checkedness, since T4 is likewise a checked tone but surfaces as *H* under right-dominant sandhi, whereas T5 surfaces as *L*.

6.2.3 The issue of T5 tetrasyllables

The long-domain sandhi forms of T5 complicate the picture of tone melodies in ways we have not addressed. As shown in § 2.2, Table 3, Xu and Tang (1988, p. 24) report two tetrasyllabic patterns for T5. In our discussion of tone melodies and melody reversal (Figure 2, § 4.3) we labeled these ‘T5a’ and ‘T5b’. Pattern T5b mimics the pattern for T3 tetrasyllables exactly.

- (19) 洛阳大学 ‘Luoyang University’
 $/lo\gamma^{T5} \text{ j}\tilde{a} \text{ da } fio\gamma/$
 T3: $lo\gamma^L \text{ j}\tilde{a}^H \text{ da}^M \text{ fio}\gamma^{ML}$

(20) 扬州大学 ‘Yangzhou University’
 $/j\tilde{a}^{T3} \text{ ts}\gamma \text{ da } fio\gamma/$
 T5a: $j\tilde{a}^L \text{ ts}\gamma^L \text{ da}^L \text{ fio}\gamma^{LM}$
 T5b: $j\tilde{a}^L \text{ ts}\gamma^H \text{ da}^M \text{ fio}\gamma^{ML}$

The fact that these patterns are reported to exist poses a challenge to our position that there is no H tone in the underlying form of T5. We argued in § 4 that the exact reason T1–T4 melodies need to end with the signature *M–ML* sequence and T5 melodies don’t is because T5 lacks a High tone while T1–T4 all have one. Examples like T5b in (20) seem to be a clear refutation of that claim.

There are, however, two reasons not to assign these long-domain forms the same evidential weight as the shorter T5 paradigm, nor to view them as a challenge to the proposal that T5 lacks an underlying High tone. In fact, we submit, it could even be interpreted as evidence to the contrary. In short: the emergence of *L–H–M–ML* pattern in T5 polysyllables is due to analogical leveling, i.e., the disappearance or erosion of an irregularity or distinction due to the influence of a more frequent, regular pattern replacing an infrequent and/or irregular one. (Cf. English *shoon, eyen, bretheren* > *shoes, eyes, brothers*.)

Word-level tone-sandhi domains exceeding three syllables in length are themselves very uncommon in Shanghainese. In a quantitative analysis of 1,206 polysyllabic domains in conversational material, Takahashi (2019b) found that two- and three-syllable domains vastly outnumbered longer ones. (See Table 26.) More than 90% of the polysyllabic domains in Takahashi’s sample were two or three syllables long, while only 8.6% contained four or more syllables.

Length (in σ)	Tokens	%
2	796	66.0%
3	306	25.4%
4	85	7.0%
5	18	1.5%
6	1	0.08%

Table 26: Distribution of polysyllabic tone-sandhi domains in Takahashi (2019b).

The rarity of long domains is compounded by the lexical rarity of T5 itself. A count of distinct character-tone pairs in the Shanghainese portion of WikiHan (Chang et al., 2022) yields only 224 T5 entries out of 2,791 (8%), making T5 the least frequent of the five tone classes in that lexical sample. A rough count of the homophone table in Xu and Tang (1988) produces the same ranking, with T5 again accounting for only about 7% of the entries. These figures are lexical-type rather than token frequencies, but they reinforce the point that long T5-initial domains draw on a particularly small portion of the lexicon, precisely the kind of marginal environment in which analogical remodeling would be unsurprising.

Note also that the two tetrasyllabic T5 patterns are nowhere near equally distributed. Takahashi (2015) compared the T5a pattern (which he calls ‘Final Association’) with the T5b pattern (his ‘Second Association’), as well as a third possibility in which a four-syllable expression is parsed as two separate disyllabic sandhi domains. His production study tested four morphosyntactic structures and found an unmistakable asymmetry: T5a occurred in every structure tested, whereas T5b occurred only in 2+2 noun compounds, i.e., four-syllable symmetric compounds of two disyllabic nouns, as in (19) and (20). Indeed, apart from tokens classified as errors, the asymmetric noun compounds and structures containing particles were realized categorically with T5a. The results are summarized in Table 27.

According to Takahashi (2015) (citing Qian, 1995, p. 323), the T5a paradigm is attested in Old Shanghainese (mostly spoken by speakers born in the first half of the 20th century), whereas the T5b pattern developed only later. Takahashi proposes that, because the ‘Second Association’ pattern, $(X-)H-(M-)ML$, was present in the tone melodies of every tone class in the language other than T5, “speakers regarded Second Association as the ‘unmarked’ sandhi variant in Shanghai Chinese and began extending it to T5 sandhi” Takahashi (2015, p. 67).

These historical facts not only defuse the threat that the T5b pattern seemed to pose to our position that T5 lacks an underlying High tone, but actually strengthen it. After all, if every other

Structure	T5a pattern	T5b pattern	Two-domain parse
	<i>L-L-L-LM</i>	<i>L-H-M-ML</i>	
2+2 noun compound	13 (24%)	33 (60%)	9 (16%)
3+1 noun compound	51 (100%)	0	0
2+1 noun + particle	55 (100%)	0	0
3 noun + particle	51 (100%)	0	0

Table 27: Distribution of T5 realizations across morphosyntactic structures in Takahashi (2015, p. 64). Percentages are those reported by Takahashi; a small number of tokens classified by Takahashi as mispronunciations are omitted.

tone class of Shanghainese came to bear the signature *M-ML* ending, why then would T5 be the lone holdout? If T5 were underlyingly identical to T3, this would be quite difficult to explain. On our view, it is not: T5 is the odd one out because it is the only one without an underlying High tone.

6.2.4 A fallback analysis

Taken together, we believe the facts above make /LM/ the most justifiable candidate for the underlying representation of T5, independently of the help that that assumption lends to our account of the phonological behavior the various tone melodies.

At the same time, the empirical coverage of our analysis does not rest entirely on this particular assumption. If the reader is unconvinced, and if the distinction between the final targets of T3 and T5 is rejected, the triggering condition expressed by $\text{SHIFT-R}(H)$ could instead be reformulated over an independently motivated, albeit less simply defined, class. Under the three-parameter classification employed by Rose (1990) and Zhu (1995), T2–T4 constitute the non-falling tones that are either long or upper, while T5 is both lower and short. The relevant class could therefore be characterized as follows:

(21) Alternative shifting class: $\{\text{long} \cup \text{upper}\} \cap \text{non-falling}$

Tone	Register	Truncation	Contour
T1	upper	long	falling
T2	upper	long	non-falling
T3	lower	long	non-falling
T4	upper	short	non-falling
T5	lower	short	non-falling

Table 28: Hypothetical alternative characterization of the T2–T4 class, using the three-parameter classification of Rose (1990) and Zhu (1995).

A shifting requirement restricted to this class would preserve the relevant empirical distinction between T2–T4 and T5. It would do so, however, at the cost of replacing a simple generalization over a shared right-edge tonal target with reference to a conjunction of register, truncation, and contour properties. While we are partial to our original, purely tone-based formulation of $\text{SHIFT-R}(H)$, it could be revamped in terms of the alternative generalization, and

thus our broader analysis of the sandhi system does not stand or fall with our representation of T5 as /LM/.

6.3 Other possible grammars

6.3.1 Ranking possibilities

In our analysis above we were forced to make conclusions about certain ranking relations based on the veracity of the predictions they made. In a few cases, we assumed a particular order for convenience. Figure 6 shows the possible ranking relations that would yield identical results to the ones we got earlier. The dotted lines mean to show that $\text{ANCHOR-L}(t)$ must be dominated by at least one of the two constraints $\ast\text{MELODYREVERSAL}$ and $\ast\text{CONTOUR}$.

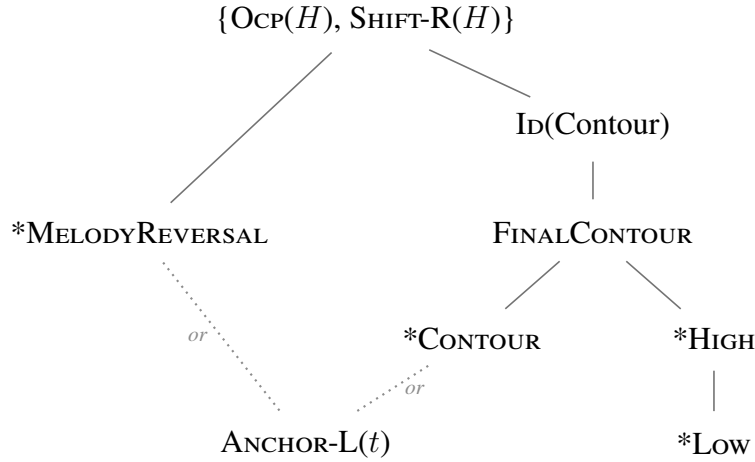


Figure 6: Complete, disjunctive ranking hierarchy. The disjunction indicates that $\text{ANCHOR-L}(t)$ must be dominated by at least one of MELODYREVERSAL and CONTOUR ; either ranking relation, or both, yields the same predictions.

6.3.2 An analysis with alternative constraints

It may be interesting to note that the last two constraints we mentioned— $\ast\text{MELODY REVERSAL}$ and $\ast\text{CONTOUR}$ —could actually be replaced. We could instead use a constraint that simply punished movement across tonal distance somewhat in the spirit of Flemming (2001). The constraint formalizes the generalization that phonetic distance is costly, and the cost of tonal distance over the course of a melody is cumulative.

- (22) $\ast\text{STEP}(t)$: Assign one violation for every step of tonal distance between adjacent tonal targets in the output melody. The total number of violations is:

$$\sum_{i=1}^{n-1} |t_i - t_{i+1}|$$

where n is the number of tonal targets in the output melody and t_i is the numerical pitch value of the i th target.

The constraint $\ast\text{STEP}(t)$ assigns violations according to the distance traversed between adjacent tonal targets. Given the three-level scale $L = 1$, $M = 2$, and $H = 3$, movement between

adjacent levels incurs one violation, while movement from one end of the scale to the other incurs two. (E.g., $L-L$ incurs no violation, $H-M$ incurs one, and $H-L$ incurs two, etc.) Violations are cumulative across the melody. For example, the sequence $H-M-L$ incurs two violations, one for each step, while $H-L-M$ incurs three: two for the movement from H to L and one more for the subsequent movement from L to M .

Note that if we made this replacement, the constraint would have to be ranked relatively low, certainly below $\text{ANCHOR-L}(t)$ and FINALCONTOUR , so that, e.g., a T3 trisyllable did not surface as $*M-H-M$ or $*M-H-ML$ instead of $L-H-ML$. This would necessitate another constraint to stop epenthesis in the case of T4 melodies, cf. $*HM-H$ versus $M-H$. This could be accomplished with a familiar DEP constraint:

- (23) $\text{DEP}(t)$: Assign one violation for every tonal target in the output in excess of the number of tonal targets present in the input.

The reader may check that the constellation of constraints in Figure 7 makes the same predictions that we made in § 4 and 5.

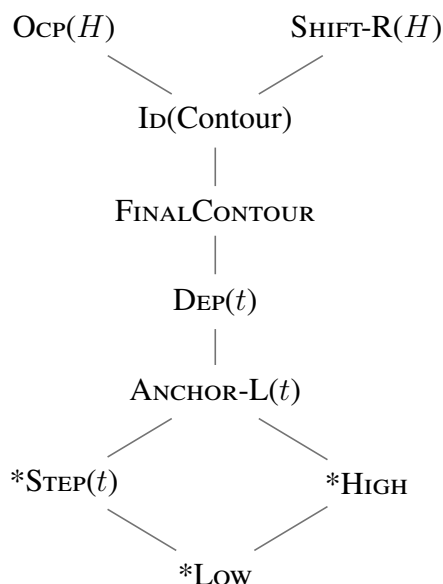


Figure 7: Alternative constraint hierarchy with $*\text{STEP}(t)$ and $\text{DEP}(t)$.

Our main reason for choosing not to advocate this alternative is that it fails to express and even clashes with what we feel are the actual reasons for why Shanghainese tone melodies have the properties they do. E.g., under an evaluation with $*\text{STEP}(t)$, the two melodies below tie:

$$M \nearrow H \searrow M \searrow L = 1 + 1 + 1 = 3$$

$$M \nearrow H \searrow M \nearrow H = 1 + 1 + 1 = 3.$$

The tie can be broken by $*\text{HIGH}$ dominating $*\text{LOW}$, but we are skeptical that this analysis—in which two consecutive steps down are as bad as one step up and another step back down again—is psychologically and physiologically as realistic as one in which the avoidance of a melody reversal is a fundamental part of the analysis.

7 Conclusion

We have argued that the word-level tone melodies of Shanghainese are best understood as a fundamentally *tonal* phenomenon. Where much previous work collapses some of the five tone classes onto a shared underlying representation and then recovers their divergent behavior from segmental, metrical, or prosodic structure, we have instead taken the traditional five-way tonal contrast at face value and let the differences among the sandhi patterns fall out of a single, uniformly ranked constraint hierarchy.

The account might equally be called a *High-tone-based* one, since the signature melody of each tone class turns on the answer to a single question: *where is the High tone?*

If the answer is “there is no High tone,” the melody stays as flat as it can, departing from the monotone of its initial target only where a final contour is forced upon it. This is T5. Long regarded as the most enigmatic of the five classes, it is in fact the most transparent: with no High tone to place, it has no reason to climb the register and no reason to fall, and it simply stays low until the last syllable demands the cheapest available rise—to Mid.

If the answer is “on the right edge,” the melody is identical to that of every other class with the same answer, save for the tonal target unique to its left edge: T3 keeps its Low, T2 keeps its Mid, and T4—whose left and right edges are one and the same High—is driven to a default Mid on its first syllable by the ban on adjacent High tones, rendering its melodies indistinguishable from those of T2. Having reached a High peak, each of these melodies must turn downward, if they continue at all; and it descend rather than reverse, closing in a falling contour rather than a rising one, because the grammar would tolerate neither a contourless final syllable nor a change of direction it can avoid.

If the answer is “on the left edge,” the melody follows the very same course as the previous three, differing only in that the lead-up to the High peak is absent: here the High simply comes first. That is T1, and its disyllabic *H–ML* form—once thought to require special machinery—needs none. It reaches its peak immediately and then finishes the only way an already-peaked, contour-final melody can.

These generalizations were formalized with a compact inventory of OT constraints. Two correspondence constraints, *ANCHOR-L(t)* and *SHIFT-R(H)*, fix the left edge and drive the rightward shift of an initial High that is the trademark of Shanghainese sandhi; a relative markedness scale over the three registers ($H > L > M$) supplies the default Mid; and a small set of markedness constraints on melodic shape—above all the prohibition on adjacent High tones (*OCP(H)*), the aversion to melodic reversal (**MELODYREVERSAL*), and the demand for a prosodic-final contour (*FINALCONTOUR*)—conspire to determine everything in between. The apparent long-distance “spreading” of T5 and the “premature” boundary tone of T1 are, on this view, not operations at all but by-products of these general pressures.

The same grammar reaches beyond the polysyllabic melodies that motivated it. With the single addition of a contour-faithfulness constraint, *ID(CONTOUR)*, it derives the monosyllabic citation forms of all five classes, including the reductions of T1 and T3 that traditional pitch-scale descriptions already record. And the one pattern that seems to defy the logic—the innovating T5b tetrasyllable, which ends in the same *M–ML* as the High-toned classes—turns out on closer inspection to reinforce it: confined to the a rare and most marginal corner of the lexicon, it is most naturally seen as a diachronic case of analogical leveling in progress toward the dominant pattern, which is not an unexpected trend if T5 is the lone class without an underlying High.

The cost of this account is a somewhat richer set of underlying representations than the most reductive analyses assume. The corresponding gain is that the divergent sandhi behavior of the five classes, their citation forms, and even their exceptions all follow from independently moti-

vated constraints on tonal correspondence and melodic shape, with no appeal to the segmental makeup of the syllable or the metrical structure of the domain. What remains open is precisely what we have set aside: how sandhi domains are delimited in the first place, how the word-level system relates to the right-dominant sandhi of the phrase, and whether the same High-tone-centered logic extends to the other Northern Wu varieties whose melodies rival Shanghainese in intricacy. We leave these questions to future work.

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